GLYCOSIDES

Topic Outline

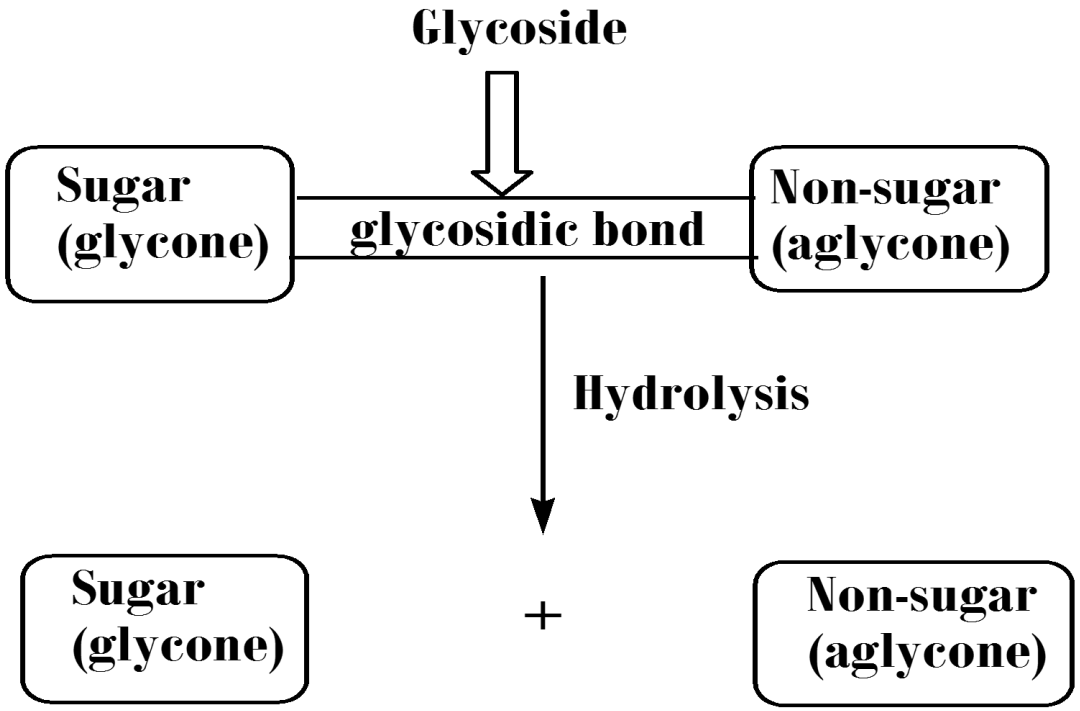
- General properties of glycosides
- Classification of glycosides
- Classes of glycosides:
  1. Anthraquinones glycosides
  2. Saponins glycosides
  3. Cardiac glycosides
  4. Simple phenolic glycosides
  5. Cyanogenetic glycosides
  6. Isothiocyanate glycosides

Glycosides

- A glycoside is an organic compound, usually of plant origin, that is composed of a sugar portion linked to a non-sugar moiety.
- The sugar portion is called glycone, while the non-sugar portion is called aglycone or genin.
- Non-sugar molecule known as the aglycone and may be
  - amino acid derivatives
  - Steroids
  - triterpenes
- The linkage between the sugar and the aglycone is an acetal linkage.

1. Introduction

- Glycosides are natural products composed of
  - Sugar (glycone) parts and
  - Non-sugar parts (aglycone or genin).



Why do we study glycosides?

- A number of glycosides exist and some are medicinally useful.
- Many glycosides are toxic
  - Many are ecologically important.
  - They could also serve as potential source of new drugs.
- Some glycosides serve as starting materials for semi-synthetic drugs.

Glycosides

- Glycosides consist of a sugar(glycone) residue covalently bound to a different structure called the *aglycone.*
- The sugar residue is in its cyclic form and the point of attachment is the hydroxyl group of the hemiacetal function.
- The sugar moiety can be joined to the aglycone in various ways:

1. Oxygen *(O-glycoside)*
2. Sulphur *(S-glycoside)*
3. Nitrogen *(N-glycoside)*
4. Carbon (*Cglycoside)*

- Modern system of naming glycosides using the termination '-oside' (e.g. sennoside).
- The term 'glycoside' is a very general one which embraces all the many and varied combinations of sugars and aglycones.
- More precise terms are available to describe particular classes. Some of these terms refer to:

1. the sugar part of the molecule (e.g. *glucoside*).
2. the aglycone (e.g. anthraquinone).
3. the physical or pharmacological property (e.g. saponin “soap-like”, cardiac “having an action on the heart”).

According to the configuration of the glycosidic linkage

1- β-glycosides:

Here the sugar involved has the β configuration at the hemiacetal carbon.

The majority of plant glycosides isolated are β-glycosides.

2- α -glycosides: Here the sugar involved has the α configuration at the hemiacetal carbon; the sugar is usually of the L-series e.g. L-rhamnose. There are only few medicinal α-glycosides known, especially the rhamnosides.

Chemical Properties of glycosides

- *O-Glycosides* can easily be cleaved into sugar and aglycone by hydrolysis with acids or enzymes. Most numerous kind found in nature
- Glycosides represent a group of natural compounds of diverse chemical structures.
- They only share the presence of one or more monosaccharide units in the structure.
- However, the following general properties should be known:

1. Glycosides do not reduce Fehling's solution, but the sugars they produce after hydrolysis reduce Fehling's solution (ppt of red cuprous oxide).

2. Glycosides show no mutarotation except after hydrolysis (C-glycosides are exceptions).
3. Most glycosides are soluble in polar solvents such as
    - water and alcohols
    - They are generally insoluble in non-polar solvents while the aglycones are soluble in ether, chloroform and insoluble in water.

The presence of sugar increases the solubility of the glycoside in water, but depends on the number the sugar units.

Chemical Properties of glycosides

- Most of the Glycosides are hydrolysable by boiling with acids
- Almost all glycosides are accompanied by enzymes which can hydrolyze them.
- The enzymes can be found in different part or cell of the plant or in the same cells being membrane bound.
- During injury to the tissue, germination or other physiological activity, the enzymes may come in contact with the glycosides and the hydrolysis of the glycosides takes place.

Physical property

- Most glycosides are crystalline compounds.
- Glycosides are generally
  - soluble in water, alcohol and hydroalcohols and
  - less soluble in nonpolar organic solvents
- The aglycone is
  - insoluble in water and
  - soluble in alcohol and other nonpolar organic solvents
- The higher the sugar content, the less soluble are glycosides in organic solvent.
- Glycosides form levorotatory solutions in water and alcohol.
- Most glycosides have bitter taste.

Pharmacological properties of Glycosides

- It is associates with the non-sugar moiety.
- The sugar part influences the stability and ultimately pharmacokinetic property of the drug in the body.
- Some glycosides are not active in their glycosidic form, but on hydrolysis they yield active aglycones. Such drugs are known as reactionary drugs.

Classification of Glycosides

1-According to the botanical origin of the producing plants.

- Example:
  - the glycosides of senna leaf are sennosides A, B, C ..etc.,
  - those isolated from Digitalis purpurea leaves are purpurea glycosides A, B, C...etc. and
  - salicin is obtained from Salix species.

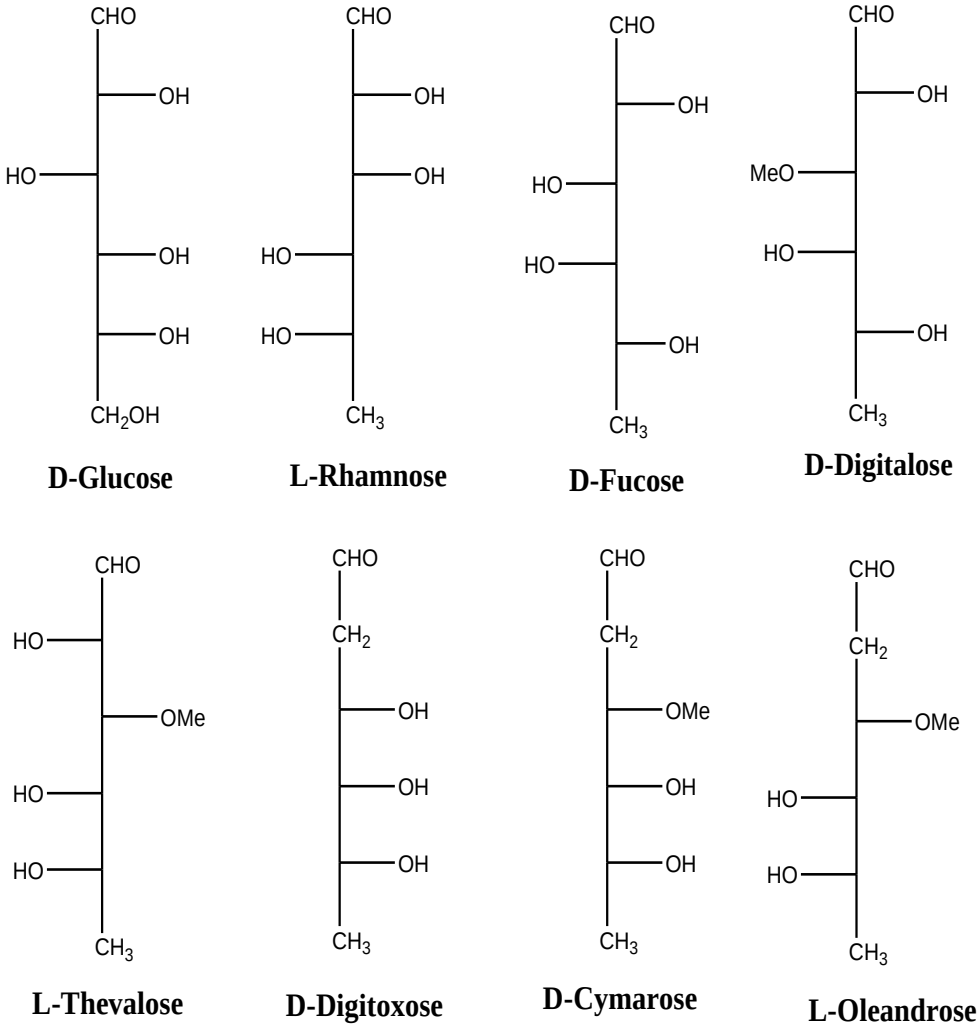
2-According to the name of the glycones present in their structures.

- For ex.
  - Glucosides is a term given to the glycosides having glucose as the sugar part, i.e. rhamnosides when the sugar is rhamnose or galactosides when the sugar is galactose and so on.

Glycosidal Sugars

- Sugars found in glycosides may be
- Monosaccharides
  - Glucose
  - Rhamnose
  - Fructose
- Deoxysugars (more rare)
  - Cymarose

  E.g. those found in the cardiac glycosides



Some sugars found in cardiac glycosides

3-Depending on the number of sugar units attached to the aglycone, glycosides may also be termed

- Monosides
- Biosides
- triosides... etc.

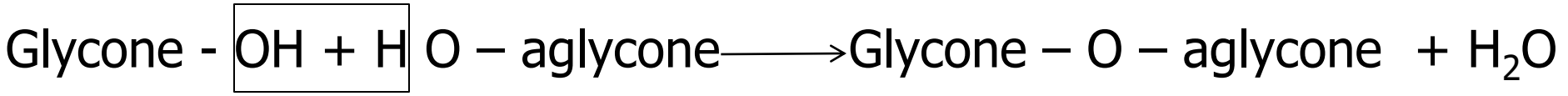
4-According to their pharmacological activities, glycosides can be named as

- cardiac glycosides
- anti-inflammatory
- anti-rheumatic glycosides
- renal disinfectants
- Analgesics
- laxative glycosides
- counter irritant etc

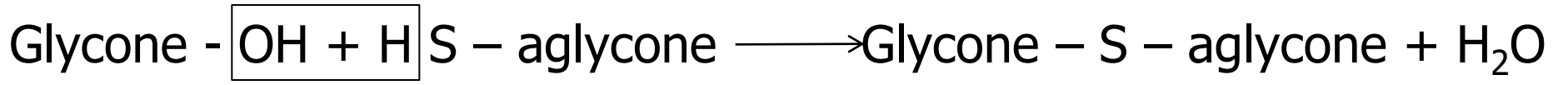
5- According to the type of the glycosidic linkage is also used to classify the glycosides into two major groups either β-glycosides or α-glycosides.

6-According to the functional group of the aglycone involved in the glycosidic linkage. i.e , there are

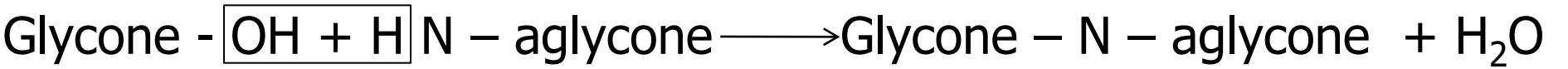
- **<u>O-Glycosides</u>**: most of the glycosides are formed by combination of a sugar(s) with the alcoholic or phenolic -OH group of a second compound (aglycone). E.g., Senna, rhubarb etc.



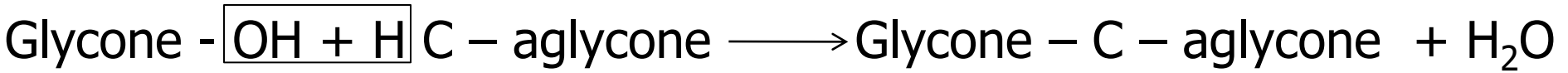
- **<u>S-Glycosides</u>**: when the linkage involves a sulfur atom of the aglycone which confirms - (thiol, SH group). E.g., Sinigrin from black mustard



– **N-Glycosides**: when the aglycone (containing -NH group) is linked to the sugar through the N atom.



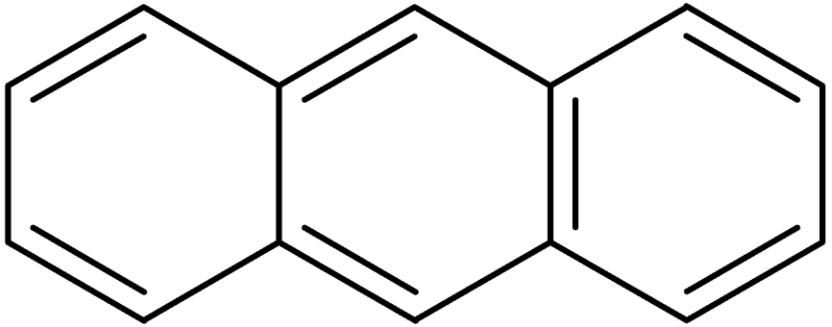
– **C-Glycosides**: where the glycosidic linkage involves a covalent bond between the sugar and the aglycone, it is a rare type of glycosides. This class of glycosides resists the usual hydrolysis procedures. E.g., Aloe



7-According to the chemical structure of the aglycone.

- Anthraquinone glycosides
- Saponin Glycosides
- Sterol or Cardiac glycosides
- Simple alcoholic and phenolic glycosides.
- Flavonoid Glycosides
- Isothiocyanate Glycosides
- Cyanogenetic glycosides.
- Coumarin glycosides.
- Alkaloidal glycosides or glyco-alkaloids
- Antibiotic Glycosides.
- Complex glycosides (tannins, polysaccharides, derived).

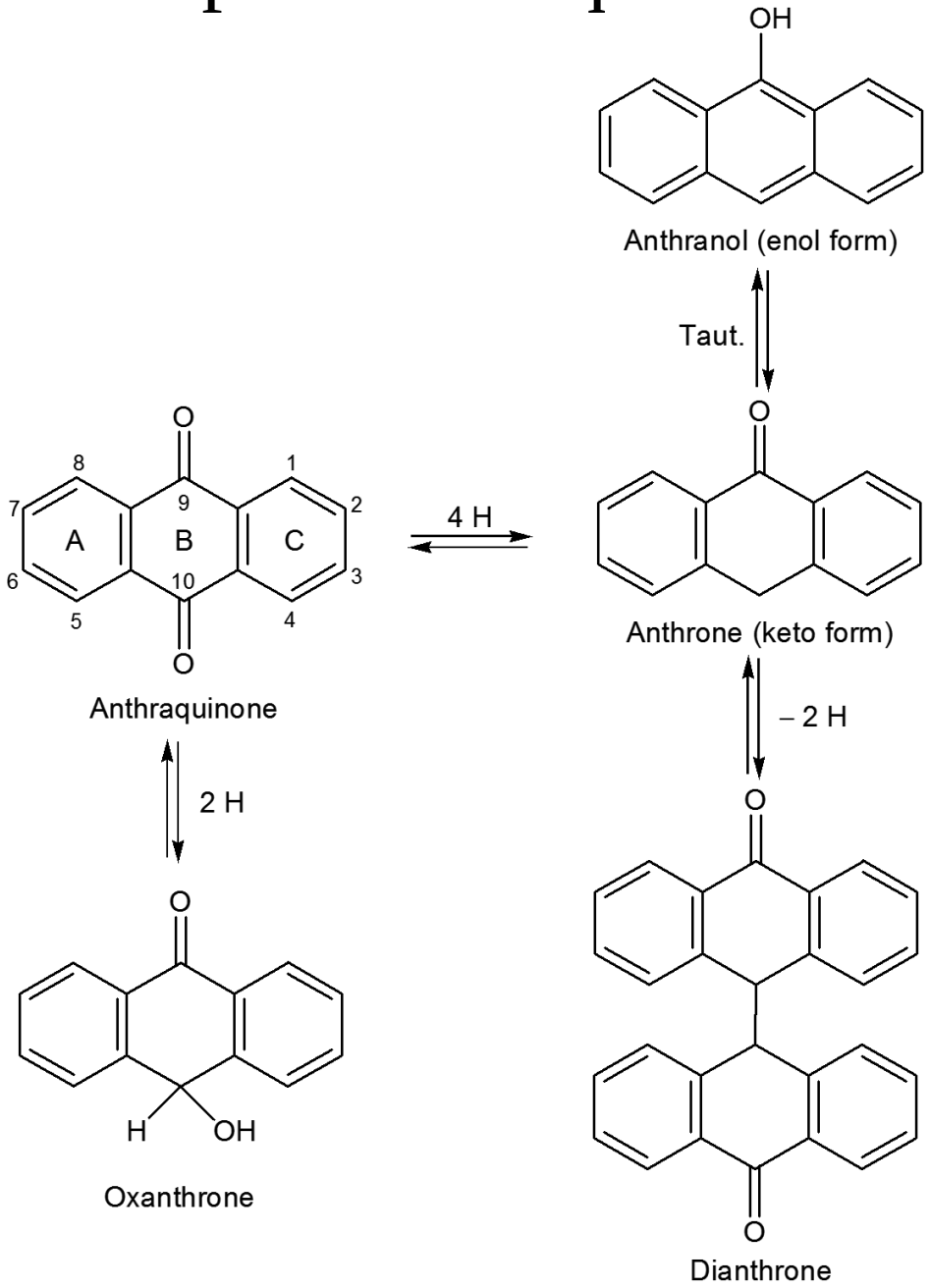
1. Anthracene glycosides



Anthracene

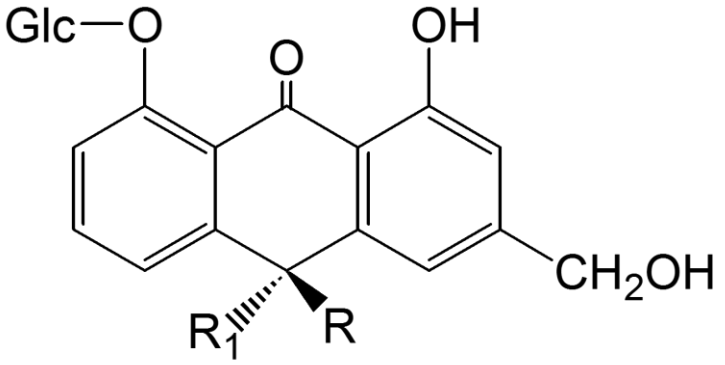
- A number of glycosides in which the aglycones are anthracene derivatives occur as the pharmacologically active constituents of several cathartics of plant origin; e.g. cascara, rhubarb, aloe and senna.
- These anthracene glycosides are sometimes referred to as the anthraquinone glycosides or the anthraglycosides.

Interrelationship of anthraquinone derivatives



- **Glycosides:**
  - **O-glycosides:**

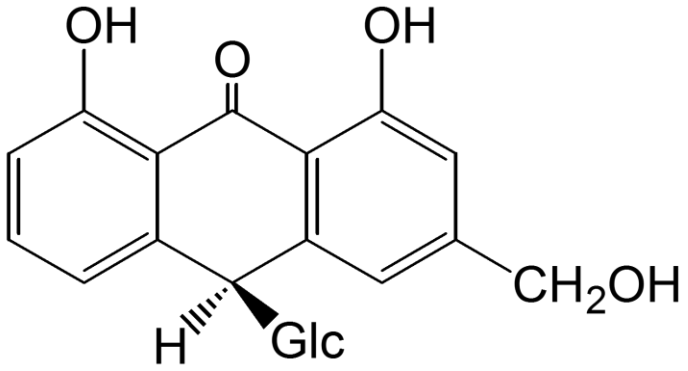
e.g. Cascarosides A & B. They are both O- and C-glycosides. Each one contain two sugar units.



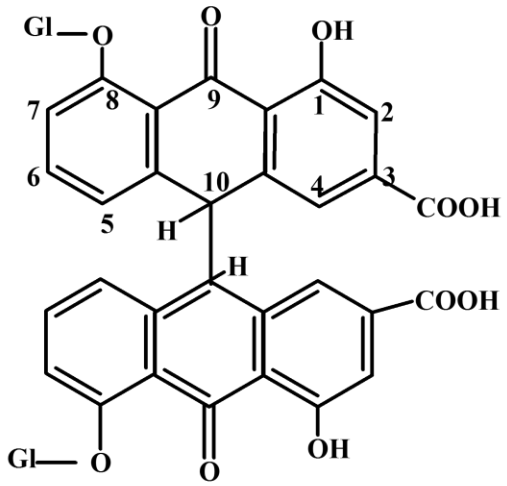
R= Glc, R_1= H — **Cascaroside A**
R= H, R_1= Glc — **Cascaroside B**

  - **C-glycosides:**

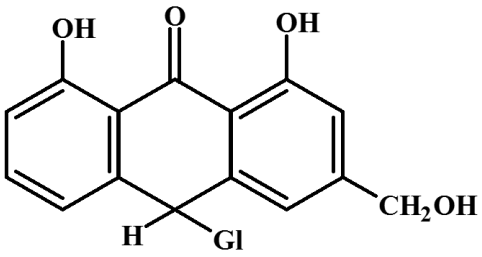
e.g. Barbaloin is formed by the removal of one sugar from Cascarosides.



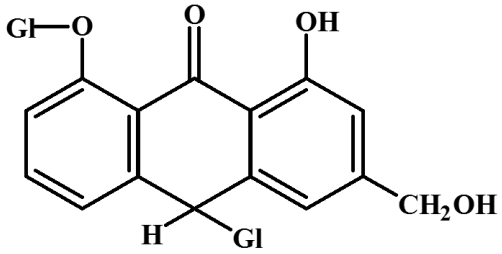
CHEMICAL CONSTITUENTS



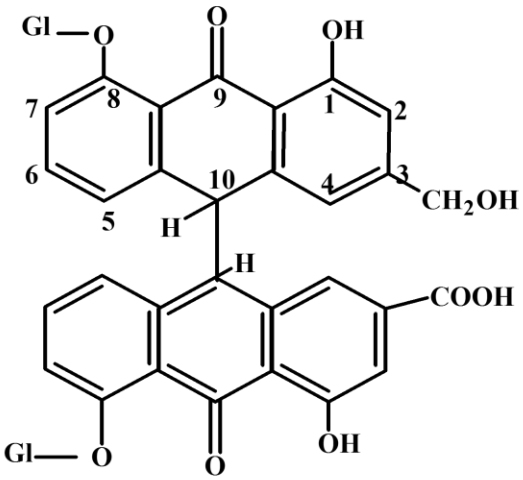
Sennosides A&B



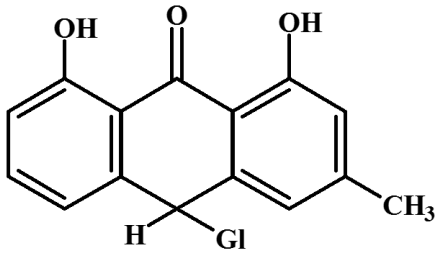
Barbaloin



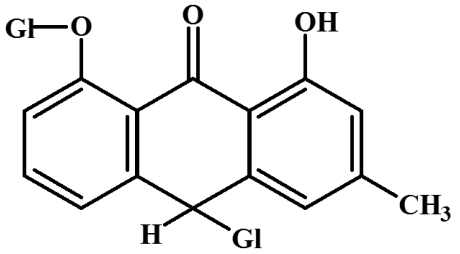
Cascaroside A& B



Sennosides C&D



Chrysaloin



Cascaroside C & D

CRUDE DRUGS CONTAINING ANTHRAQUINONE GLYCOSIDES

- **Senna**
- **Senna pods**
- **Aloes**
- **Rhubarb**
- **Cascara**

Senna

- Synonyms: (senna folium, senna-ki-patti, tinnevelly senna cassia senna)
- Biological source: It consists of the dried leaflets of *Cassia senna* (Alexandrian senna), or *Cassia angustifolia* (Tinnevelly senna).
- Family: LEGUMINOSAE.
- Geographical source:
  - Indigenous to Africa (tropical regions). Used since 9th – 10th century
  - Introduced into medicine by Arab physicians (used both the leaves and pods)
  - Exported by Alexandria – name of the Sudanese drug.

Comparison of Macroscopical characters

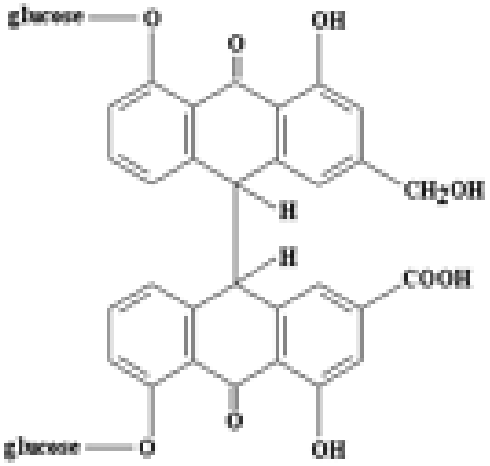
INDIAN SENNA	ALEXANDRIAN SENNA
YELLOWISH-GREEN	PALE-GREYISH GREEN
7-8mm width and25 -60mm length	2-4cm long and 7-12mm wide
broken and normally flat	Broken and curled at edges
Often shows impressions	Few press markings

Comparison of Chemical test

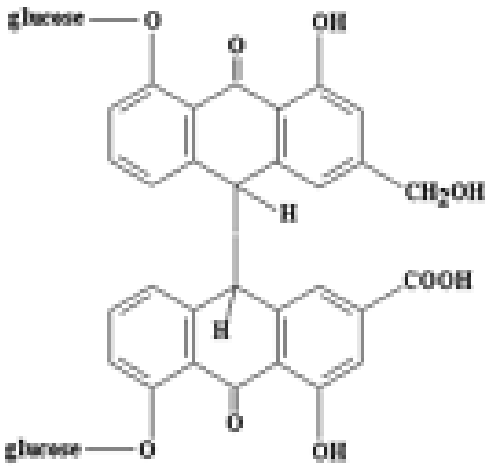
Ether extract of hydrolysed acid solution of herb with methanolic magnesium acetate solution gives	
Pink colour in daylight	Orange colour in daylight
Pale green-orange colour in filtered UV light	Yellow-green colour in filtered UV light

Chemical Constituents

- 2 active glycosides: Sennoside A
  Sennoside B
  - Both hydrolyse: 2 molecules glucose + aglycones: Sennidin A and Sennidin B.
- Sennoside C & Sennoside D
- Kaempferol (yellow flavanol)+ glucoside (kaempferin)
- Rhein
- Aloe-emodin
- Palmidin A (Rhubarb)
- Mucilage
- Calcium oxalates
- Resin



SENNOSIDE A



SENNOSIDE B

ADULTERANTS

- Tinnevelly senna and alexandrian senna are adulterated with
  - DOG SENNA (papillose cells in lower epidermis)
  - PALTHE SENNA (No Anthraquinone glycosides)
  - MECCA and ARABIANSENNA (brown-green colour, elongated)

Cascara Bark

Synonyms: (Cascara sagrada ,Sacred bark, Chittem bark, Cortex *Rhamnipurshianae*)
Biological source: Dried bark of *Rhamnus purshiana De candolle* of
Family: Rhamnaceae.

Collection: It is collected in between april-august when it is readily separates from wood.
By longitudinal incision of 5-10cm.

MACROSCOPY:

Colour	-	Externally purplish brown and Internally reddish brown
Odour	-	Nauseous
Taste	-	Bitter
Size	-	1-4mm thickness.

Broken pieces small and flat occurs as quills/channels.
Internally longitudinal striations.

Storage

- During preparation and storage the bark should be protected from rain and damp (partial extraction of constituents may occur or bark may become mouldy).
- Should be stored for at least 1 year before use (although no longer a BP requirement).
- Bark appears to increase its medicinal value upto 4 years old (stored bark)

- **Why Stored for a Year?**
  - When stored for at least a year – better tolerated by patient (less griping pains due to increased peristalsis)
  - Yet as effective as fresh bark.
- **Reason?**
  - Due to Hydrolysis and other changes that occur during storage.
  - Bitter taste of Cascara can also be reduced by treating the bark with alkali (alkali earths or MgO).

Chemical Constituents

- 4 main glycosides – Cascarosides A-D
  - Cascaroside A
  - Cascaroside B
  - Cascaroside C
  - Cascaroside D
- 2 aloins:
  - The C-linked glucose at C-10 protect anthrones from oxidation.
  - Breakdown products of Cascarosides A-D
    - Barbaloin present as a secondary glycoside in *Cascara* (derived from aloe-emodin)
    - Chrysaloin (derived from chrysopanol anthrone)

Constituents

- O-glycosides
- Derived from
  - Emodin
  - Emodin oxanthrone
  - Aloe emodin
  - Chrysophanol
- Dianthrones
  - Those from
  - Emodin
  - Aloe-emodin
  - Chrysophanol
  - Hetrodianthrones
    - Palmidin A, B and C (Rhubarb)

Substitutes

- *Rhamnus alnifolia* (too rare)
- *Rhamnus crocea* (bark is very different from official drug)
- *Rhamnus californica* (so closely related to *Rhamnus purshianus* some botanists do not consider them to be separate species).
- *Rhamnus fallax*

Uses

- Purgative
- Similar to Senna
- Normally as a tablet
- Also used on animals

ALOES

- **Synonyms**: ALOE, MUSSABBAR, KUMARI
- **Biological source**:

Dried juice of leaves of

Aloe ferox Miller,
Aloe barbadensis Miller (curacao aloes),
Aloe perryi Baker (socotrine aloes),
Aloe africana Miller and
Aloe spicata Baker (cape aloes)

- **Family**: Liliaceae.
- **Cultivation and collection**:

Cultivated by root suckers.
No water logging.
Juice is collected by a cut at leaf base.

Preparation of Cape Aloes

- Curacao aloes: Place the leaves in kerosene tins and kept in tilted position to drain out all the juice and boil on large copper pans and pour in metal containers.
- Cape Aloes :
  - They are placed in ground in basin shaped depression which is lined by goat skin /canvas for 5-6 hrs.
  - Cape Aloes are prepared from the wild plants *of Aloe ferox*.
  - Leaves are cut transversely near the base.
  - 200 leaves arranged around a shallow hole in the ground (lined with canvas or goatskin).
  - Cut ends overlap & drain into the canvas.
  - After 6hrs all the juice is collected.
  - Transferred to a drum.
  - Boiled for 4hrs on an open fire.
  - Poured into tins while hot → solidifies.
- Socotrine and Zanzibar Aloes: juice is collected in goat skin and solidified.

MACROSCOPIC CHARACTERS

CURACAO	CAPE	SOCOTRINE	ZANZIBAR
STRONG ODOUR	NAUSEOUS	NAUSEOUS	-
BROWNISH-BLACK yellow-brown – chocolate brown	BROWNISH-GREEN	BROWNISH-YELLLOW	LIVER BROWN
OPAQUE MASS	MASSES	OPAQUE PASTY MASS	-
BITTER TASTE	BITTER TASTE	BITTER TASTE	BITTER TASTE
UNEVEN FRACTURED	GLASSY FRACTURED	CONCHOIDAL	EVEN FRACTURED

➢ Chemical Constituents :

- C-glycosides
- Resins
- The major glycoside in Aloes is the C-glycoside Aloin
  - Barbaloin
  - Isobarbaloin
- Aloe-emodin

Cape Aloes: Also Contain

Aloinoside A & Aloinoside B (O-glycosides of barbaloin)

- The C-linked glucose at C-10 protect anthrones from oxidation.

➢ Uses

- Purgative
- Activity is increased when administered with small quantities of soap or alkaline salts; Carminatives moderate griping tendency.

Chemical Tests

- Unlike C-glycosides, O-glycosides of *Aloe* are not hydrolysed by heating with dilute acids or alkali.
- Can be decomposed with ferric chloride & dilute HLC - Modified Borntrager's Test – oxidative hydrolysis. Anthraquinones give a red colour when shaken with dilute ammonia.
- All Aloes give a strong green fluorescence with borax (characteristic of anthranols) - General test for aloes.

Substituents & Adulterants

- Socotrine, zanzibar, Natal aloes ,Mocha aloes with BLACK CATECHU.
- Natal: Dry extract of aloe candelabra for cape aloes.
- Mocha: Brownish black glossy fragments interior varieties.
- Curacao aloes with BLACK CATECHU , IRON, STONES.
- These are derived from the mucilage gel – parenchyma cells
- Should not be confused with aloes (juice of pericycle – juice used for laxative effect).
- Cosmetic industry (usefulness often exaggerated) - Used as suntan lotions, tonics and food additives.
- Mucilage = polysaccharide of glucomannans and pectin

RHUBARB

- Synonyms: Radix rheum, Revandchini, Rheum
- Biological source :Dried rhizomes of *Rheum emodi Wallich*,
  Rheum palmatum Linn ,
  Rheum webbianum Royle
- Family: polygonaceae

- Collection and preparation:
  - Drug obtained from wild plants which are draught resistant.
  - Propagated by seeds, rhizomes.
  - Harvested in spring season and the rhizomes are collected and cleaned & de-corticated, dried and packed in wooden boxes.

Chemical Constituents

1. Anthraquinones without a carboxyl group – chrysophanol, emodin, aloe-emodin & physcion.

   Also the glycosides of these substances.

2. Anthraquinones with a carboxyl group (rhein and its glycoside: glucorhein).
3. Anthrones and dianthrones of chrysophanol, emodin, aloe-emodin or physcoin.
4. Dianthrone glucosides of rhein (Sennosides A and B).
5. Hetrodianthrones derived from 2 different anthrone molecules: Palmidin A and Palmidin B.
6. Free anthraquinones: chrysophanol, emodin, aloe-emodin and rhein.
7. Some of the above constituents may also occur as glycosides.

Uses

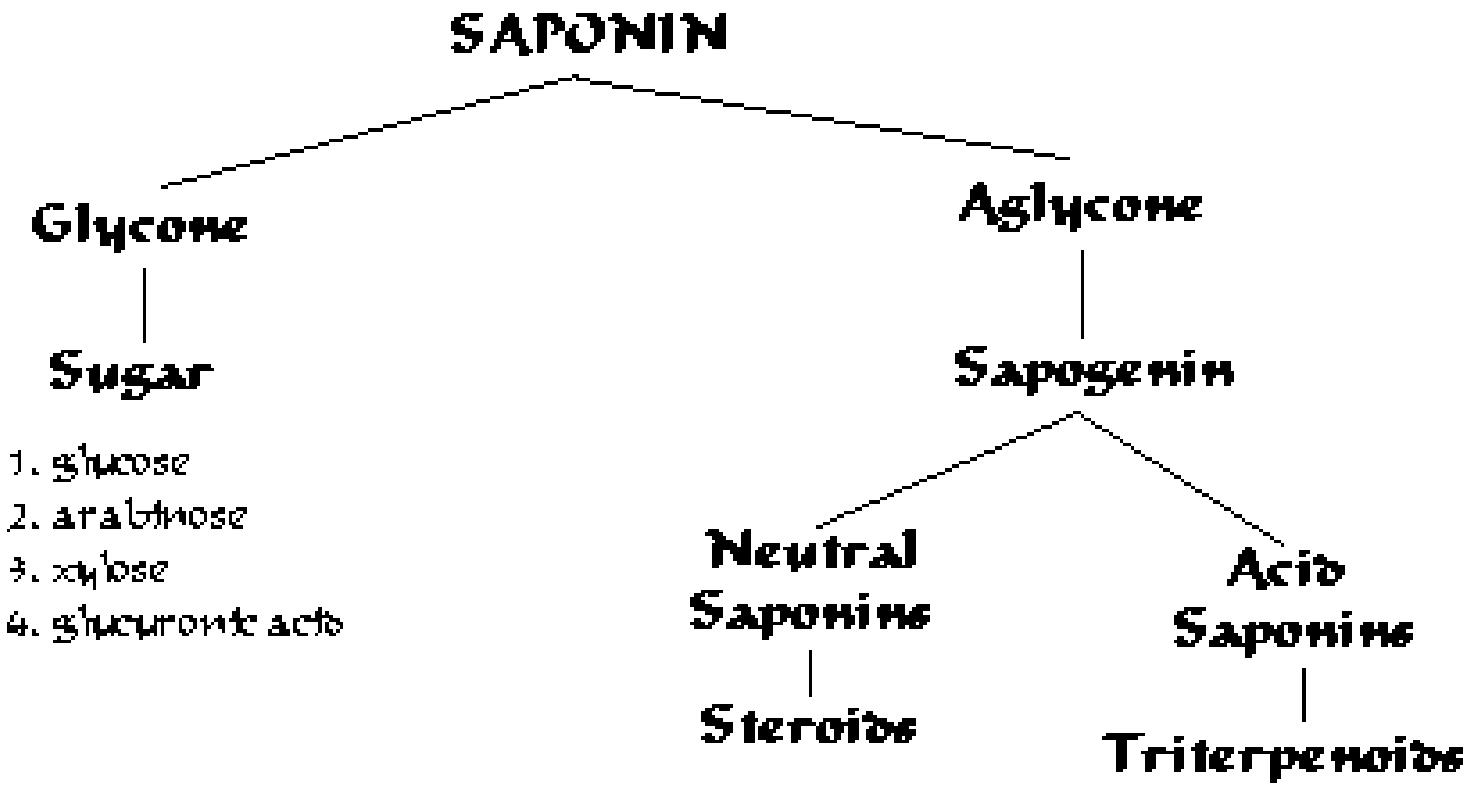
- Bitter stomachic
- Diarrhoea (low doses) – contains tannins
- Purgative (high doses) – followed by an astringent effect.
- Suitable only for occasional for occasional use, not for chronic constipation.
    - It is purgative only in large doses.
    - It contains high amount of tannins that have astringent effect.

Mechanism of Action

- The glycosides are absorbed from the small intestine and re-excreted in the large intestine where they increase the motility so produce laxation.

- Aglycones produce griping effect so it is recommended to prescribe antispasmodic with them.

2. Saponin glycosides



- A group of plant glycosides known as saponins share in varying degrees, two common characteristics:

(a) They foam in aqueous solution.

(b)They are generally considered as haemotoxic, because they cause haemolysis of red blood cells(erythrocytes). Due to this activity some of them are used as fish poisons.

The actual cause of the haemolysis:

The red blood cells carry sterols in their membranes, and when brought into contact with saponins, the sterols of the RBCs are precipitated and the colloidal chemical properties of the membrane are so altered as to give hemoglobin passage to the surrounding medium.

- The aglycones of the saponins are collectively referred to as ***Sapogenins*.**
- The more poisonous saponins are often called ***Sapotoxins*.**
- Plant materials containing saponins have long been used in many parts of the world for their detergent properties for example,
  - In Europe, the root of *Saponaria officinalis* (Fam. *Caryophyllaceae*) and
  - In South America, the bark of *Quillaia saponaria* (Fam. *Rosaceae).*
- Such plants contain a high percentage of the glycosides known as saponins (Latin *Sapo*, means Soap) which are characterized by their property of producing a frothing aqueous solution.

Properties:

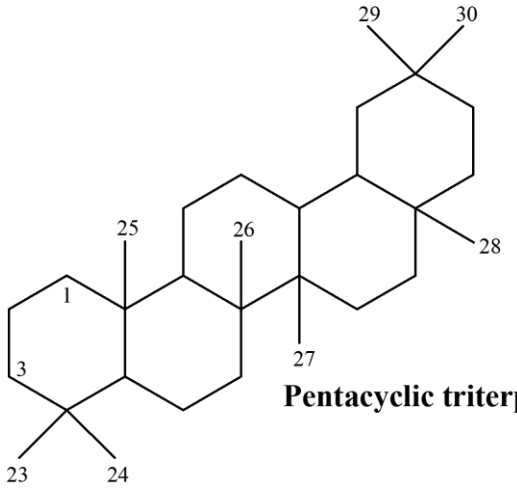
- Saponins form colloidal solution in water (hydrophilic colloids) which froths upon shaking.
- They cause irritation of mucous membrane.
- These substances modify and lower the surface tension and therefore foam when shaken.
- Use to increase the foaming of beer.
- Industrial applications of saponins include their use in cleaning industrial equipment and fine fabrics
- As powerful emulsifiers of certain resins, fats and fixed oils.

- **Structure of Saponins:**

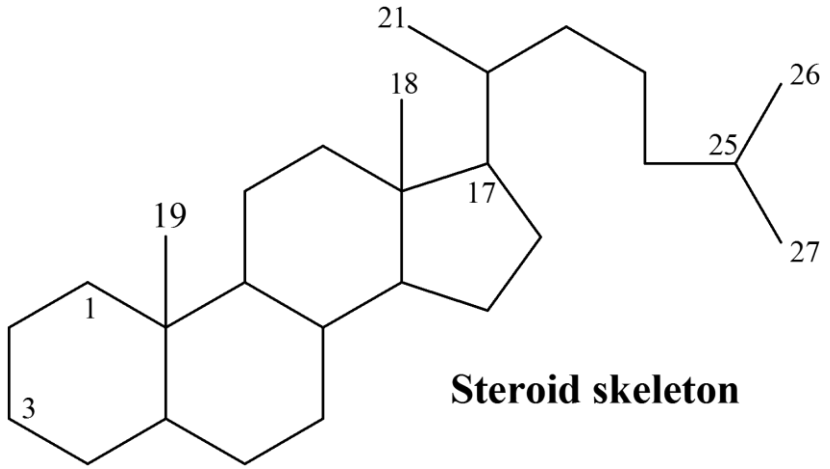
  According to the structure of the aglycone or sapogenin, two kinds of saponin are recognized:

1. The steroidal type (commonly tetracyclic triterpenoids, C-27).
2. The triterpenoid type (pentacyclic triterpenoids, C-30).

- Both of these have a glycosidal linkage at C-3 and have a common biosynthetic origin via mevalonic acid and isoprene units.

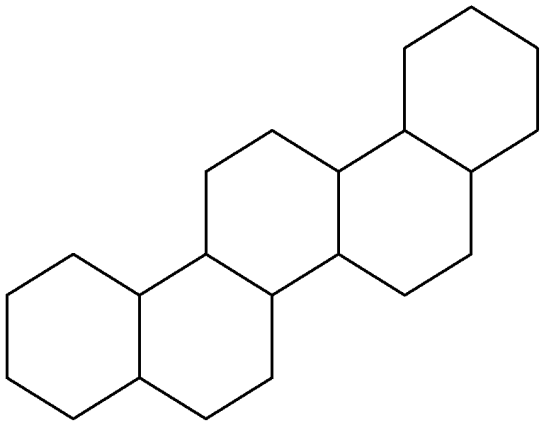


Pentacyclic triterpenoid skeleton

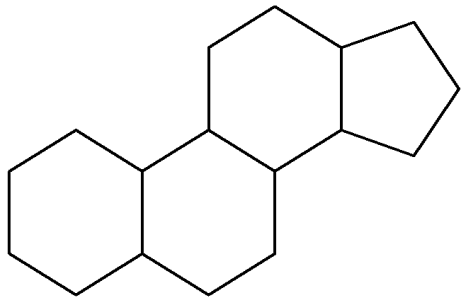


Steroid skeleton

Sapogenins



Pentacyclic skeleton



Steroidal skeleton

- Some examples of saponins
  - Dioscorea
  - Liquorice
  - Shatavari
  - Ginseng
  - Senega
  - Quillaia
  - Gokhru

A. Steroidal saponins

- The steroidal saponins are less widely distributed in nature than the pentacyclic triterpenoid type.
- Steroidal saponins are of great pharmaceutical importance because of their relationship to compounds such as the sex hormones, cortisone, diuretic steroids, vitamin D and the cardiac glycosides.
- Examples: Diosgenin (*Dioscorea sylvatica*), Sarsapogenin (*Smilax* sp.).

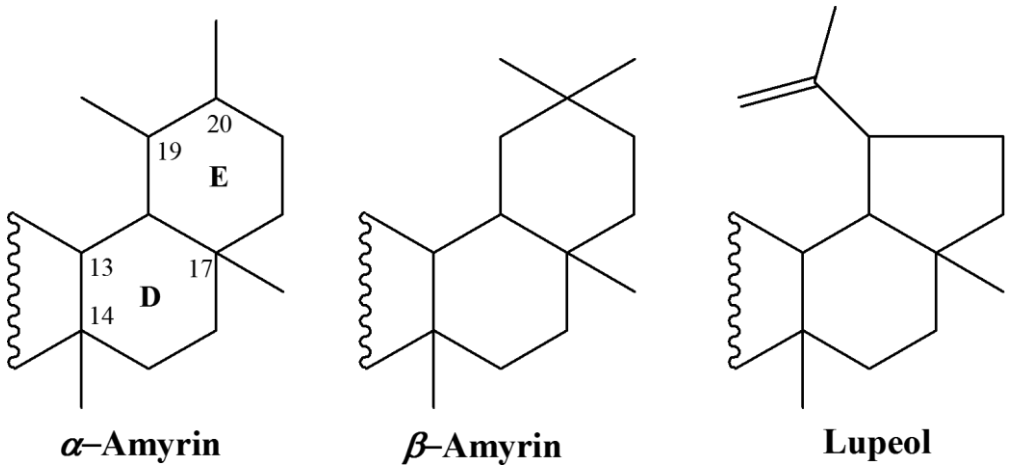
Dioscorea

- Synonyms: Yam, rheumatism root
- Biological source: It consists of dried tubers of the plants *Dioscorea deltoidea*
- Family: Dioscoreaceae
- Macroscopic characters:
- Colour - Slightly brown
- Odour - Odourless
- Taste - Bitter

- Chemical constituents:
  - Dioscorea rhizomes contain diosgenin a steroidal saponin(4 to 6%).
  - Starch 75%
  - Enzyme sapogenase
- Uses:
  - The rhizomes are used as rich source of diosgenin.
  - It is used as a precursor for synthesis of several corticosteroids, sex-hormones and oral contraceptives.
  - Used in treatment of rheumatoid arthritis.

B. Pentacyclic triterpenoid saponins

- Triterpenoid saponins my be classified into three groups represented by α-amyrin, β-amyrin and lupeol.
- Examples: Primulagenin (*Primula* sp.), Quillaiac acid (*Quillaia saponaria*) and Glycyrrhetinic acid (*Glycyrrhiza* sp.).



Liquorice

- Synonyms: Liquorice root, glycyrrhiza
- Biological source:
  - It consists of dried, peeled and unpeeled root and stolon of *Glycyrrhiza glabra* Linn.
- Family: Leguminosae
- Macroscopy:

Colour	-	Unpeeled yellowish brown Peeled is pale yellow in colour
Odour	-	Faint and characteristic
Taste	-	Sweet
Shape	-	Cylindrical pieces

- Chemical constituents:
  - The Chief active constituents of liquorice is triteropenoid saponin known as glycyrrhizin(glycyrrhizic acid), which is a potassium and calcium salt of glycyrrhizinic acid.
  - Glycyrrhizinic acid is glycoside and on hydrolysis yields glycyrrhetinic acid(glycyrrhetic acid), which has a titerpenoid structure.
  - Glycyrrhizinic acid is 50 times sweeter than sugar (sucrose).
  - Upon hydrolysis, the glycoside loses its sweet taste and is converted to the aglycone glycyrrhetinic acid plus two molecules of glucuronic acid.
  - Flavonoids (liquiritin and isoliquiritin)

- Uses:
  - Traditionally it is used as an expectorant and demulcent.
  - It is used in cough mixtures, and as a flavouring agent in formulations with nauseous drugs.
  - Flavonoids are used in peptic ulcer in the form of deglycyrrhized liquorice.
  - It is also used as antispasmodic(due to flavonoid isoliquiritin).
  - As a flavouring agent for chewing tobacco and snuff tobacco.

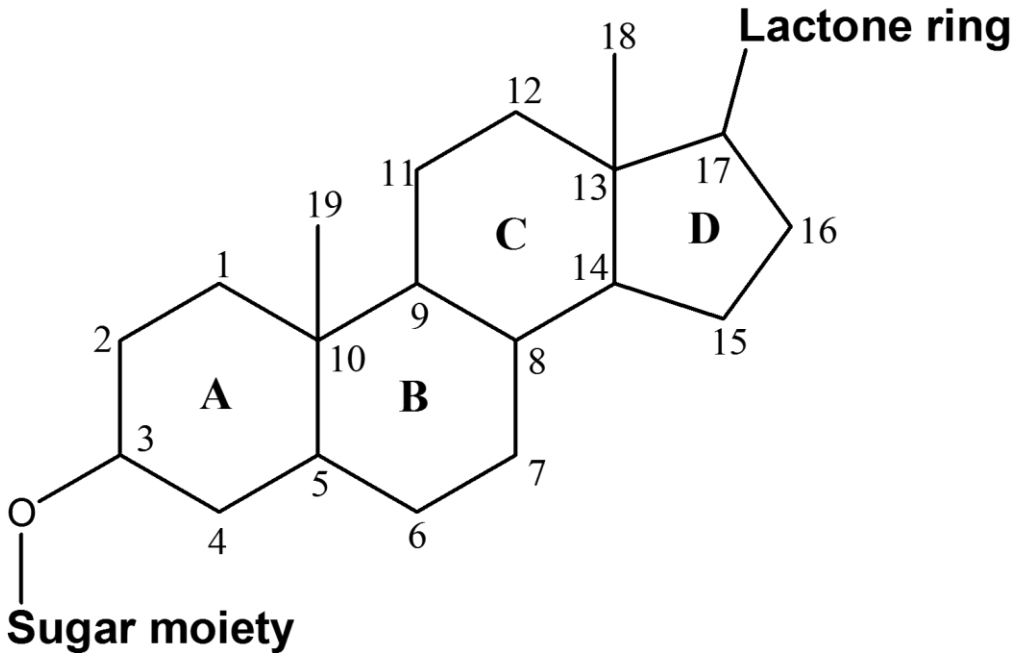
3. Cardiac glycosides

- Cardiac glycosides are a group of natural products characterized by their specific effect on myocardial contraction and atrioventricular conduction.
- In large doses they are toxic and bring about cardiac arrest in systole, but in lower doses they are important drugs in the treatment of *congestive heart failure*.
- They have a diuretic activity. Since, the improved circulation tends to improve renal secretion, which relieves the edema often associated with heart failure.

🌍 Distribution in nature

- **Cardiac glycosides** occur in small amounts in the seeds, leaves, stems, roots or barks of plants of wide geographical distribution, particularly of the family
  - *Apocyanaceae*
    - *e*.g. seeds of *Strophanthus*, roots of *Apocynum* and fruits of *Acokanthera);*
  - *Scrophulariaceae*
    - e.g. leaves of *Digitalis sp.*,
  - *Liliaceae*
    - e.g. scales of the bulbs of *Urginea* and *Convallaria*, and
  - *Ranunculaceae*
- Cardiac glycosides are also found in animals only in exceptional cases: Bufadienolides occur in toads (*Bufo*).

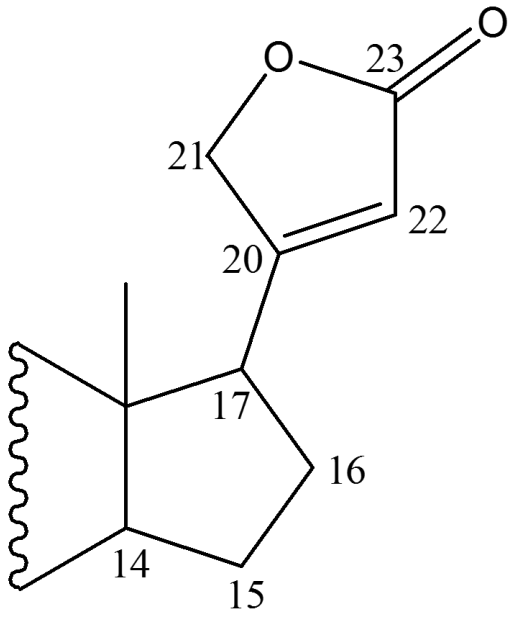
Structure of cardiac glycosides



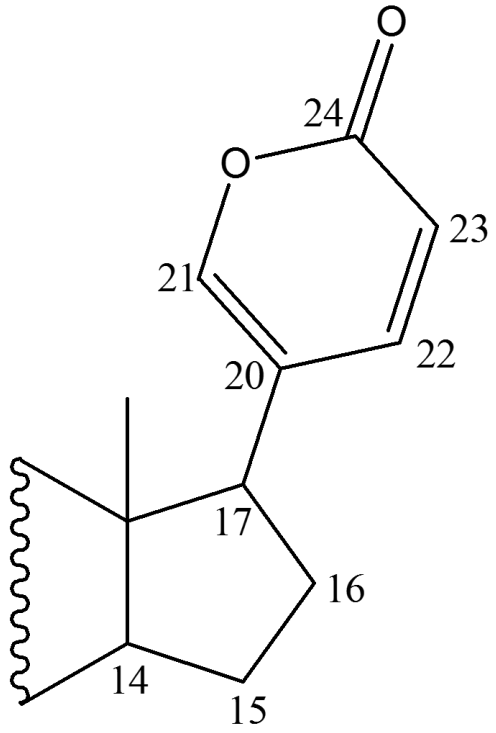
- The structure comprise a steroidal aglycone of the (C_{23}) **cardenolide** type or of the (C_{24}) **bufadienolide** type, and a sugar moiety, most often an oligosaccharide.

A. Structure of the aglycones

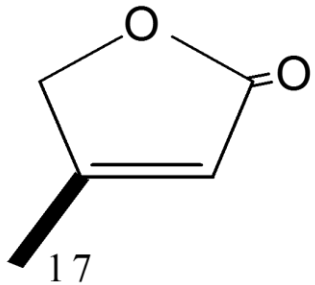
- All of the aglycones have in common the classic, tetracyclic, steroidal nucleus.
- The A, B, C and D rings normally have a *cis*-*trans*-*cis* configuration or less often, a *trans*-*trans*-*cis* configuration.
- Also common to all the aglycones is the presence of two hydroxyl groups: one is a 3β secondary alcohol, the other is a 14β tertiary alcohol.
- All of the aglycones have a β constituent at C-17: an α,β-unsaturated lactone.

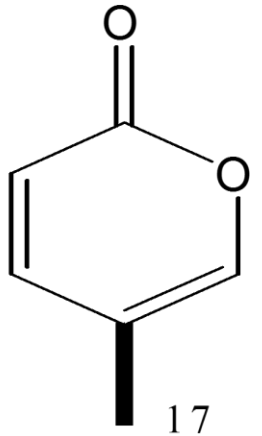


Lactone ring of
Cardenolide



Lactone ring of
Bufadienolide





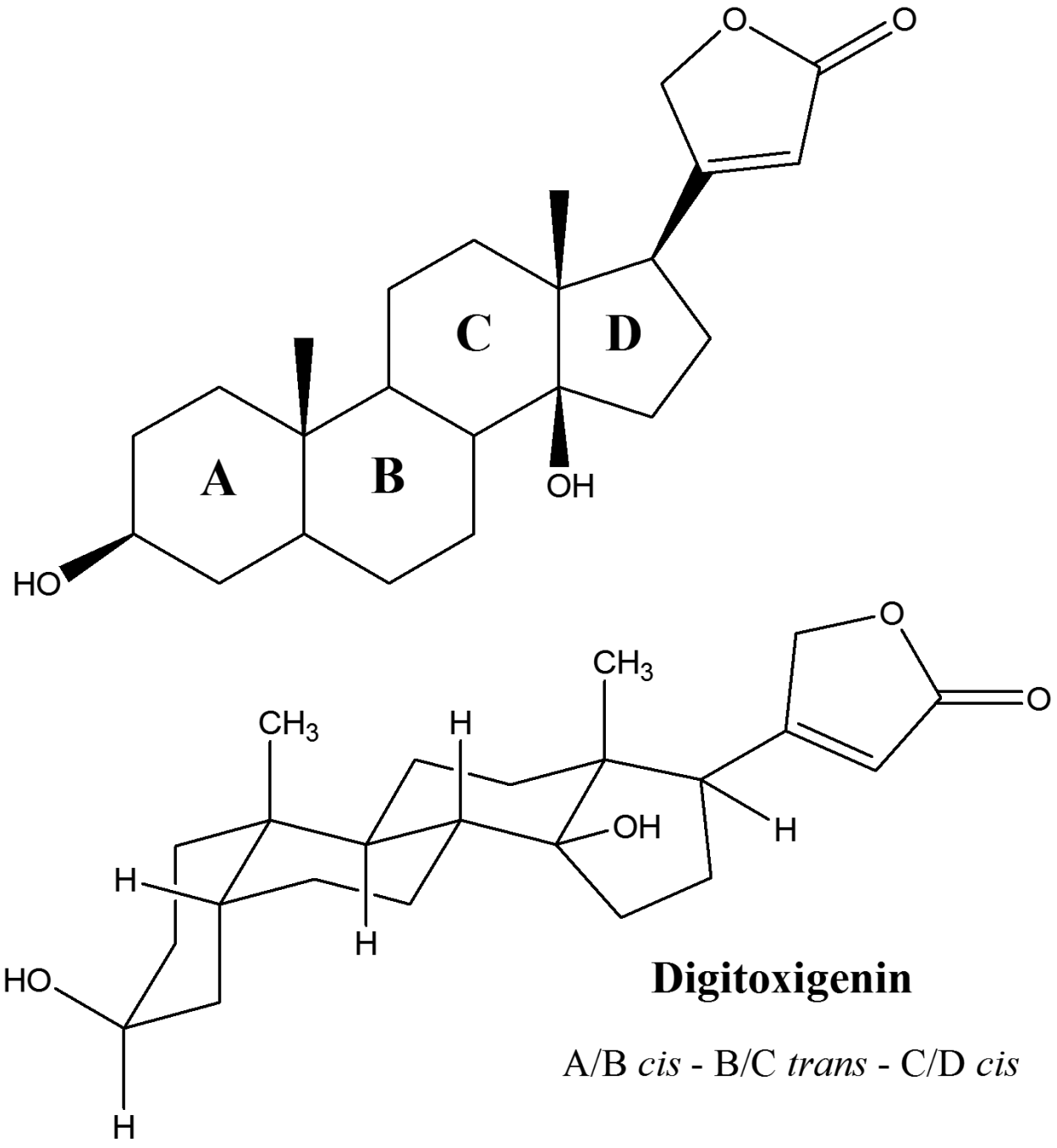
- The size of the lactone ring distinguishes two groups of aglycones: the $\mathbf{C_{23}}$ cardenolides with an α,β-unsaturated γ-lactone (= butenolide) and the $\mathbf{C_{24}}$ bufadienolides with a di-unsaturated δ-lactone (= pentadienolide).

B. Structure of the sugar moiety

- The sugar moiety is generally linked to the aglycone through the hydroxyl group at C-3.
- The majority of the saccharides found in cardiac glycosides are highly specific:

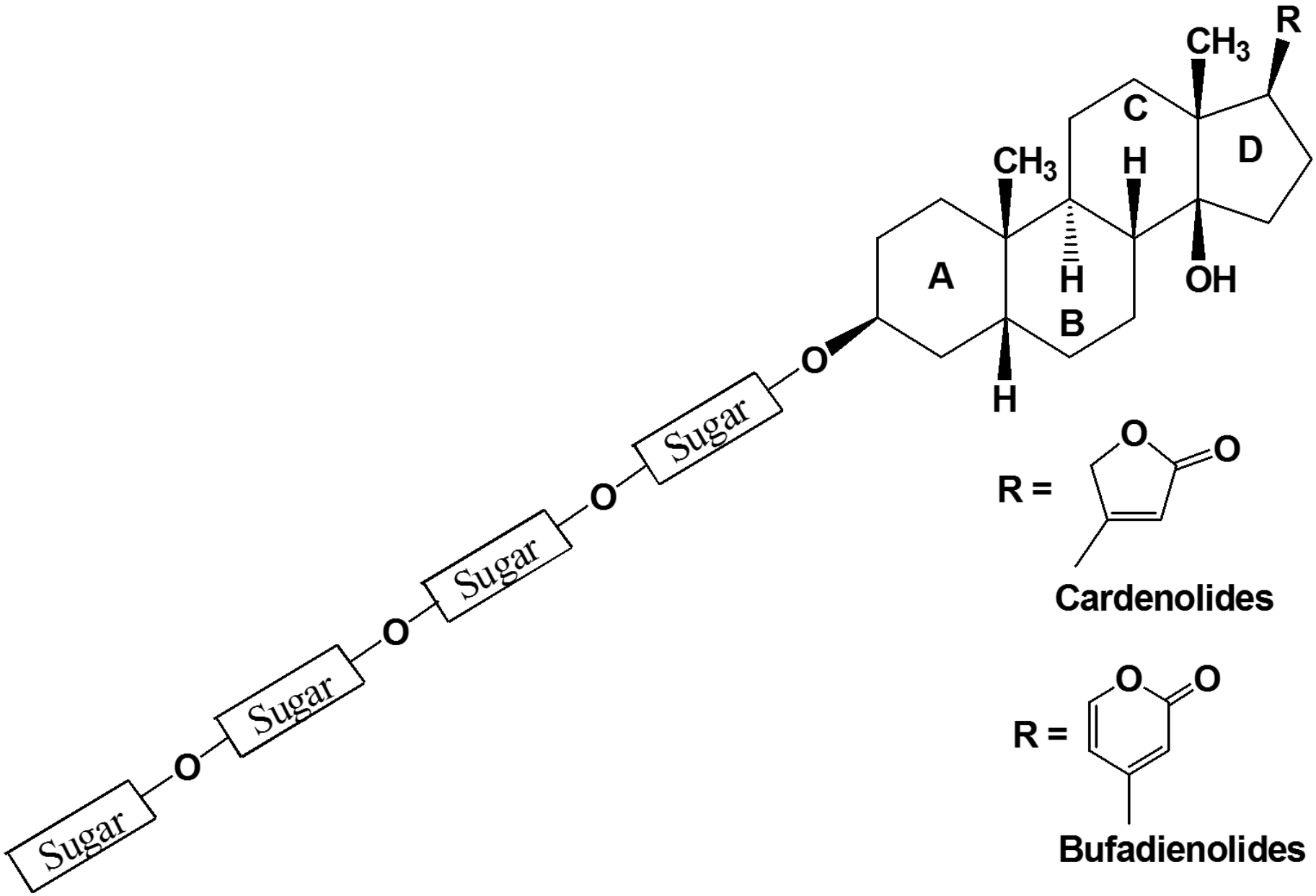
1. 2,6-dideoxyhexoses, e.g. D-digitoxose
2. 2,6-dideoxy-3-methylhexoses, e.g. D-diginose
3. 6-deoxyhexoses, e.g. L-rhamnose
4. 6-deoxy-3-methylhexoses, e.g. D-digitalose
5. Hexose, e.g. glucose (*when these is a glucose unit, it is always terminal*).

- The sugars can modify the activity (potency, toxicity), the solubility, the diffusion through membranes, the rate of absorption and transportation of the glycosides.



Digitoxigenin

A/B *cis* - B/C *trans* - C/D *cis*


Sugar
O
Sugar
O
Sugar
O
Sugar
O
A
B
C
D
H
H
H
CH3
CH3
OH
R
R =
O
O
Cardenolides
R =
O
O
Bufadienolides

Biosynthetic origin

- Aglycone of the cardiac glycosides are derived from mevalonic acid but the final molecules arise from a condensation of a C_{21} steroid with a C_2 unit (the source of C_{22} and C_{23}).
- Bufadienolides are condensation products of a C_{21} steroid and a C_3 unit.

Pharmacological properties

- Cardiac glycosides increase the force and speed of contraction of the heart. In patients with cardiac insufficiency, this positive inotropic effect translates into
  - an increase in cardiac output.
  - an increase in cardiac work capacity without any increase in oxygen consumption.
  - a decrease in heart rate, and, indirectly.
  - a decrease in arterial resistance.

Pharmacological Action of Cardiac Glycosides

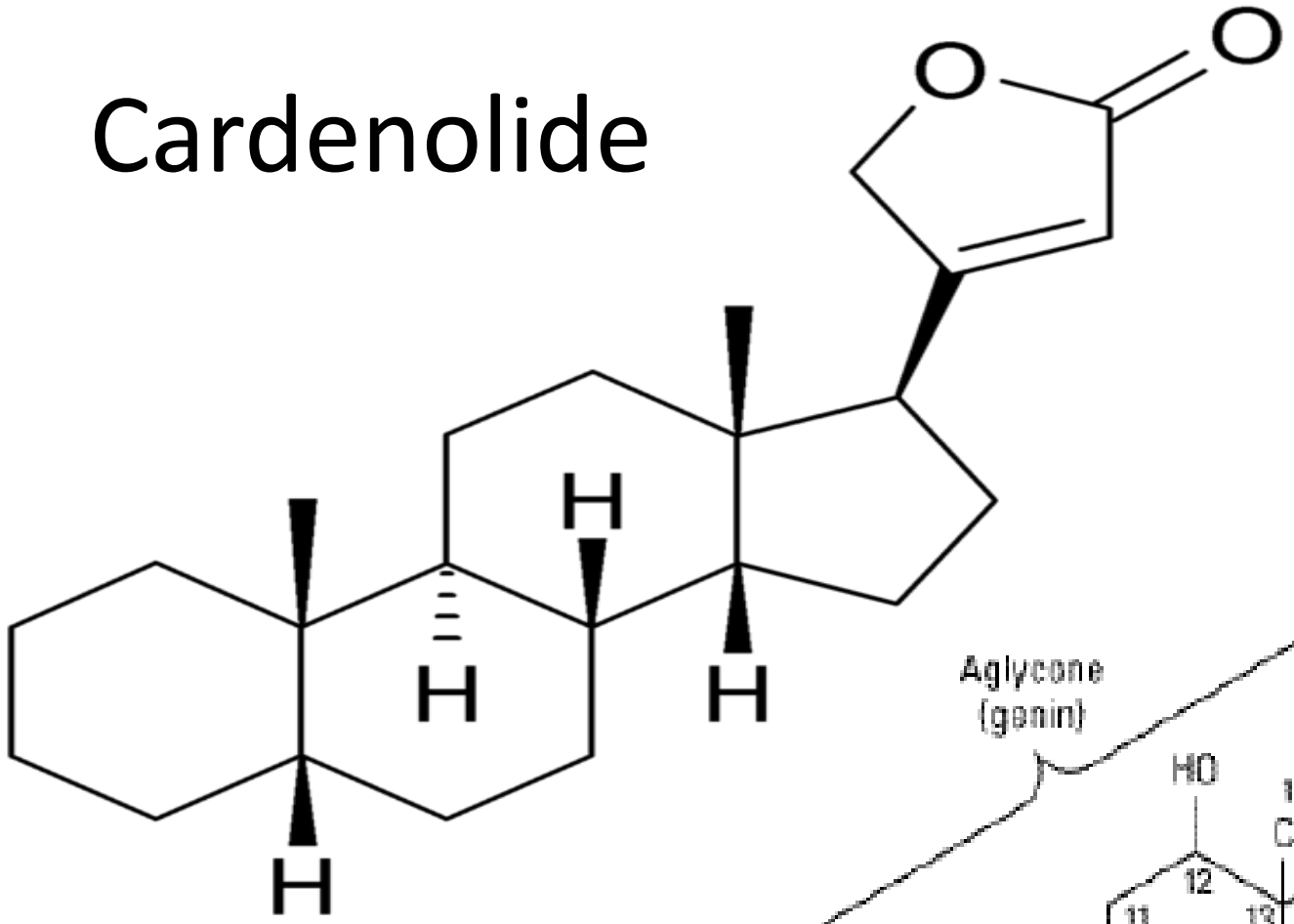
- Effectiveness depends on both the aglycones and the sugar attachments.
- Medicinal action depends on the aglycone
- But the sugars make the compound more soluble in increases the fixation of the glycoside to the heart muscle.
- It is thought to act in competition with K ions for specific receptor enzymes (ATPase) sites in the cell membranes of the heart muscle when there is an influx of Na ions.
- Effect is to increase the force of heart contraction
- Diuretic action relates to the improved circulatory effects.

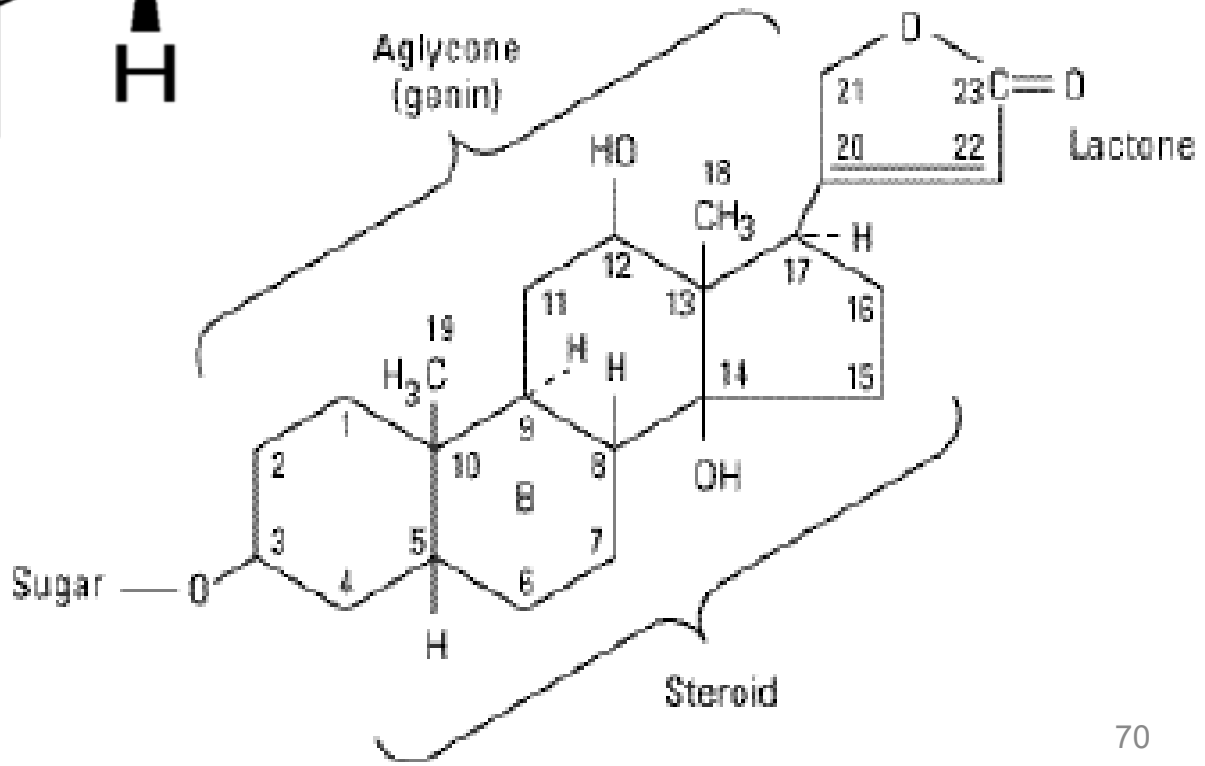
Cardiac glycosides

Influence of sugar units on absorption, on set of action and half-life of cardiac glycosides

Number of sugars	Water solubility	Route of administration	On set of action	Elimination
Less	Less	Oral	Slow	Slow
More	More	Parentral (IV)	Very fast	Fast

Cardenolide





- **Digitalis glycosides**

  Several species of *Digitalis* yield pharmacologically active principles. The most important of these species are *Digitalis purpurea* and *Digitalis lanata*.

1. Digitalis purpurea folium (Red foxglove leaves)

   0.15% – 0.4% total cardenolides, ~ 30 glycosides Purpurea glycosides A and B (~60%), digitoxin (~12%), gitoxin (~10%) and gitaloxin (~10%).

2. Digitalis lanata folium (White foxglove leaves)

   0.5% – 1.5% total cardenolides, ~ 60 glycosides Lanatosides A and C (~50%), lanatosides B, D, E as well as **digoxin** and **digitoxin**.

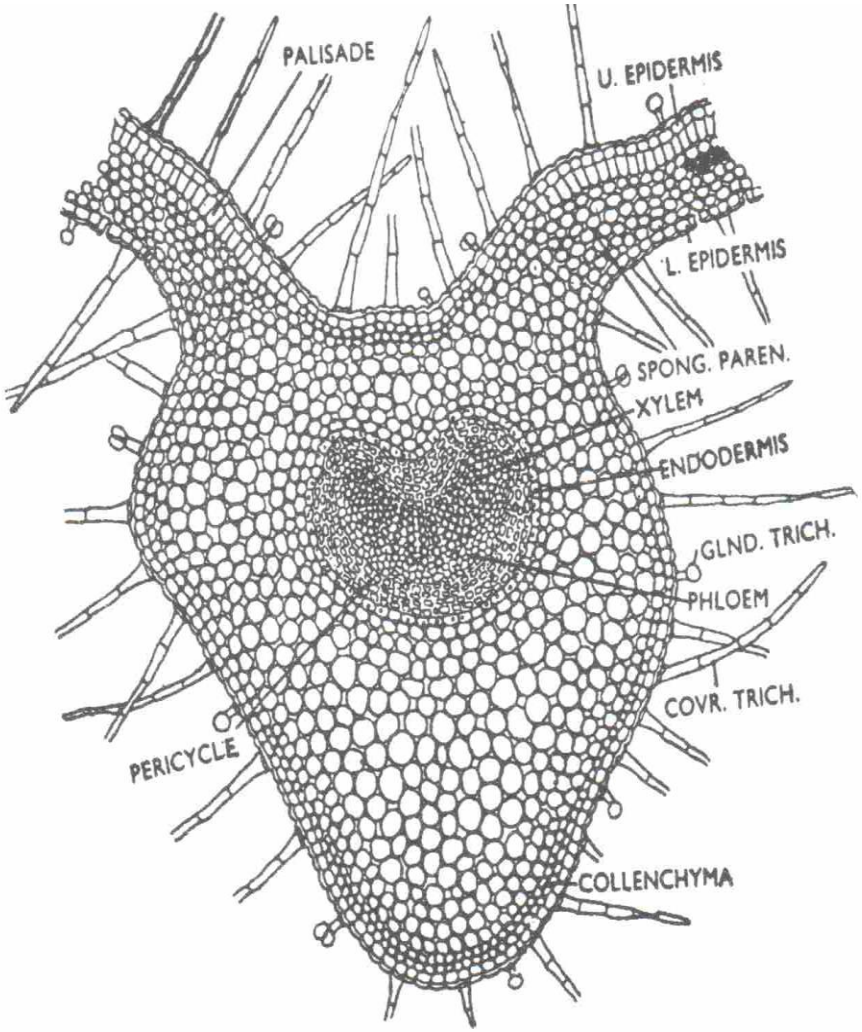
Digitalis (Purple Foxglove)

- **Biological Source:**
  - *Digitalis* consists of the dried leaves of *Digitalis purpurea*.
- Family: Scrophulariaceae
- **Habitat:**
  - Native to Western Europe. Although the plant is cultivated, wild plants are thought to be superior.
- **History:**
  - The poisonous nature of the leaves are well known, and the drug was recommended for use since 1542.
- Discovered by William Withering, an 18th century country doctor (cardiac effect)
- His work led to the production of a life-saving medicine

- It is generally agreed that the first-year leaves collected in Jan-Feb (Southern Hemisphere) have the highest content of total glycosides. After that they decrease during the winter months. After that another peak is reached during the time of flowering, but not as high as the first one.

Microscopical Characteristics

- Stomata and hairs present on both surfaces (more on lower surface)
- No calcium oxalate
- Clothing hairs are uniseriate
- Glandular hairs



Digitalis purpurea Leaf. T. S. (X 50)

Active Constituents

- Glycosides
  - purpurea glycoside A
  - purpurea glycoside B
- (2 main actives in the fresh leaf)
  - Glucogitaloxin
- At C-3 of the genin: a linear chain of 3 digitoxose sugar moieties terminated by glucoseOn drying: enzyme degradation takes place → loss of the terminal glucose. Produces
  - Digitoxin
  - Gitoxin
  - Gitaloxin
- (main actives in dried leaves)
- Poor storage leads to further hydrolysis and complete loss of action

Digitalis – Active Constituents

- also contains **anthraquinone glycosides**
- **Saponins**
- **Sopogenins**
- **Flavonoids**
- Other glycosides (present in smaller quantities)
  - **Digitalose**
  - **Glucose**
  - **Verdoxin** (x3 more toxic than gitaloxin)

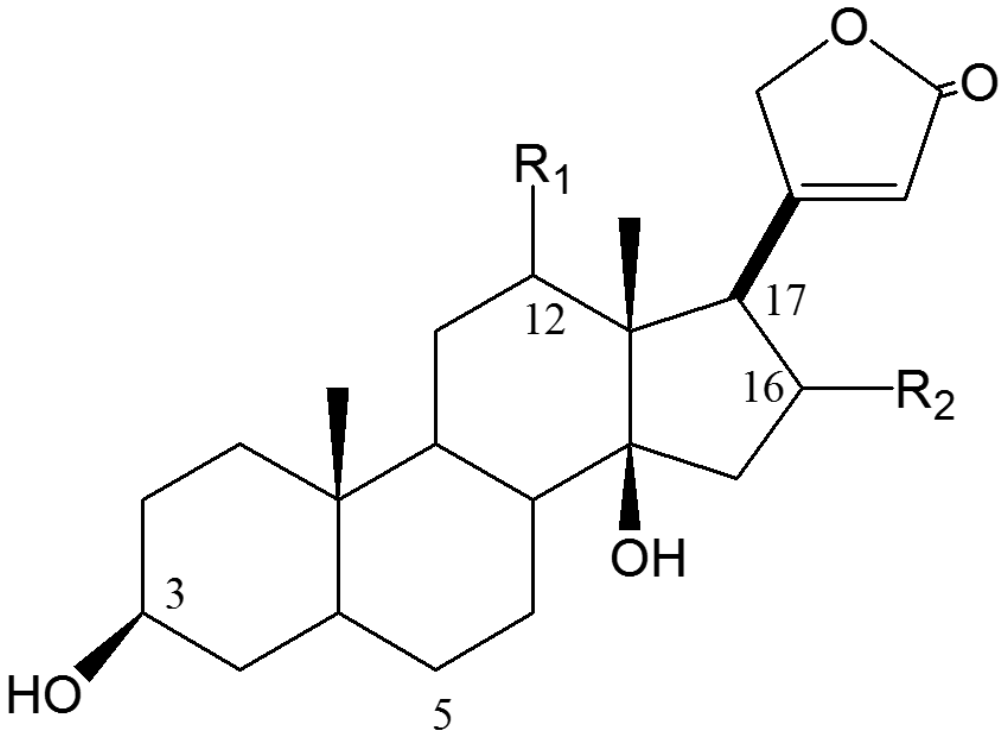
Digitalis – Medicinal Actions

- Tonic effect on the diseased heart
- Glycosides enable the heart to beat more strongly, slowly and regularly, without using or needing more O_2.
- Stimulates urine production → lessening the load on the heart.
- Other uses of the plant (and glycoside) include being used as **arrow poisonings**.

Cautions

- Potential for overdose
- Only under professional supervision
- Plant is subject to legal restrictions.

Digitalis contain three major aglycones



$R_1=R_2=H$	Digitoxigenin
$R_1=H,\ R_2=OH$	Gitoxigenin
$R_1=OH,\ R_2=H$	Digoxigenin

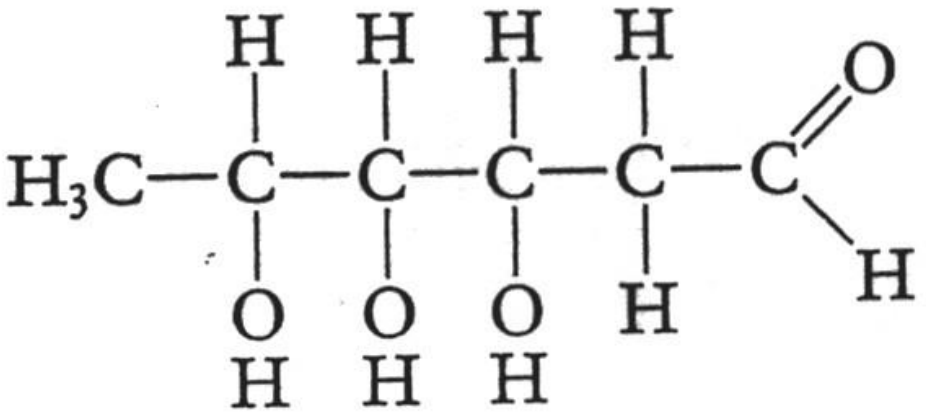
Sugars in digitoxin

2 molecules of digitose

1 molecule of 1-acetyl digitose

1 molecule of glucose

Digitose



Strophanthus

- Biological source:
  - It consists of dried seeds (ripe) of the plant *Strophanthus kombe*
- Family: Apocynaceae
- Habitat: East Africa.
- Greek *strophos*

  (a twisted cord or rope) *anthos* (a flower)

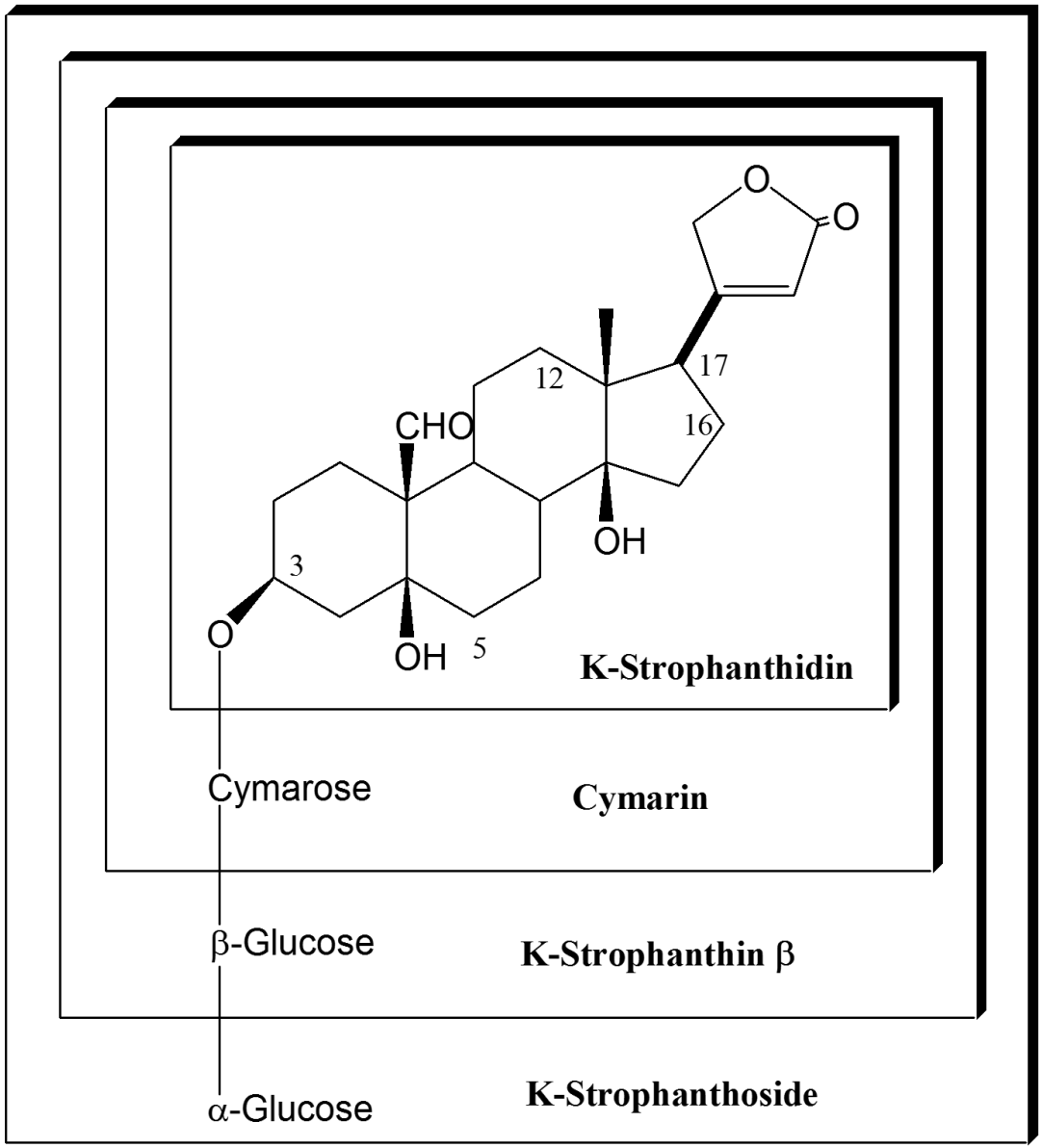
Chemical constituents:

- Strophanthoside
- strophanthin
- Cymarin
- all based on genin – strophanthidin)
- Alkaloid: Inoeine
- Fixed oil, resin
- Mucilage

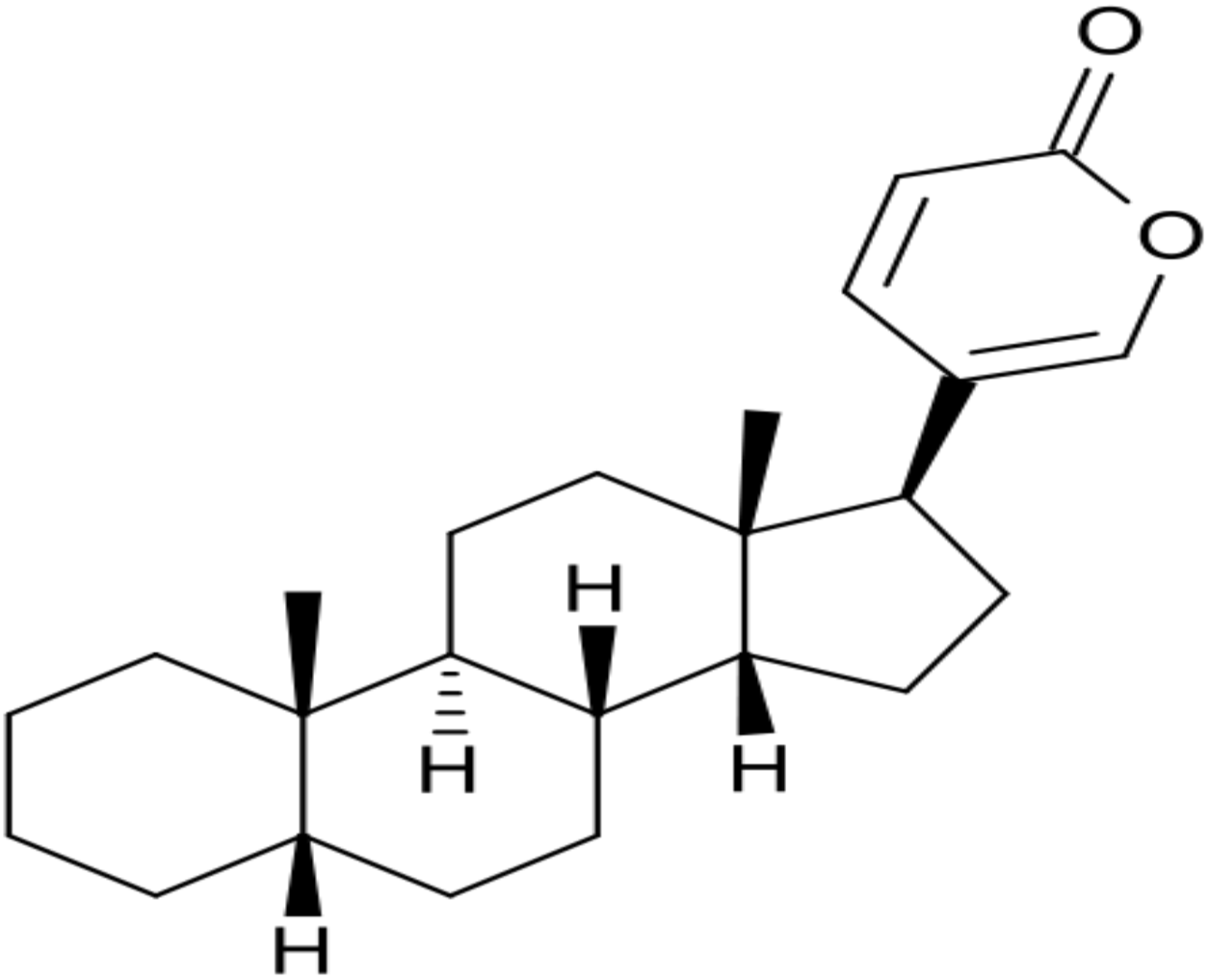
Uses - similar to *Digitalis*

- Chronic cardiac weakness
- Can be administered IV
- Poisonous
- Diuretic action

(thought to be more Powerful than *Digitalis*)


O
O
17
12
16
CHO
OH
3
O
OH
5
K-Strophanthidin
Cymarose
Cymarin
β-Glucose
K-Strophanthin β
α-Glucose
K-Strophanthoside

Bufadienolide



Bufadienolides

- Less commonly distributed in nature than cardenolides
- Occur in some Liliaceae and Ranunculaceae Species. Also occur in toad venoms.
- Therapeutically there is not much value as the therapeutic index is low and production of side effects high.
- Squill, however has a time-honoured place as an expectorant
- Widely used in the treatment of cough.

Squill

- **Biological source :**
  - Consists of the dried sliced bulbs of *Urginea maritime*, from which the membranous outer scales have been removed.
  - It is commercially known as white squill.
- **Family:** Liliaceae
- History: Squill appears in Egyptian Ebers papyrus (1500BC)
- In Greece it was used by Pythagoras & Hippocrates in the 6th & 5th centuries BC.
- **Description:**
  - A perennial herb growing up to 1.5 m from a large white (or red) bulb.
  - It has a single flowering stem, a rosette of large basal leaves, and a dense spike of white flowers.

- **Squill glycosides**

  Urginea maritima (L.)

  0.1% – 2.4% total bufadienolides, ~15 glycosides
- **White variety:** average 0.2%-0.4%

  proscillaridin A, scillaren A, glucoscillaren A (aglycone: scillarenin)

  scilliphaeoside, scilliglaucoside
- **Red variety:** < 0.1%

  scilliroside and glucoscilliroside (aglycone: scillirosidin); proscillaridin A and scillaren A as in the white variety

Actions & Uses

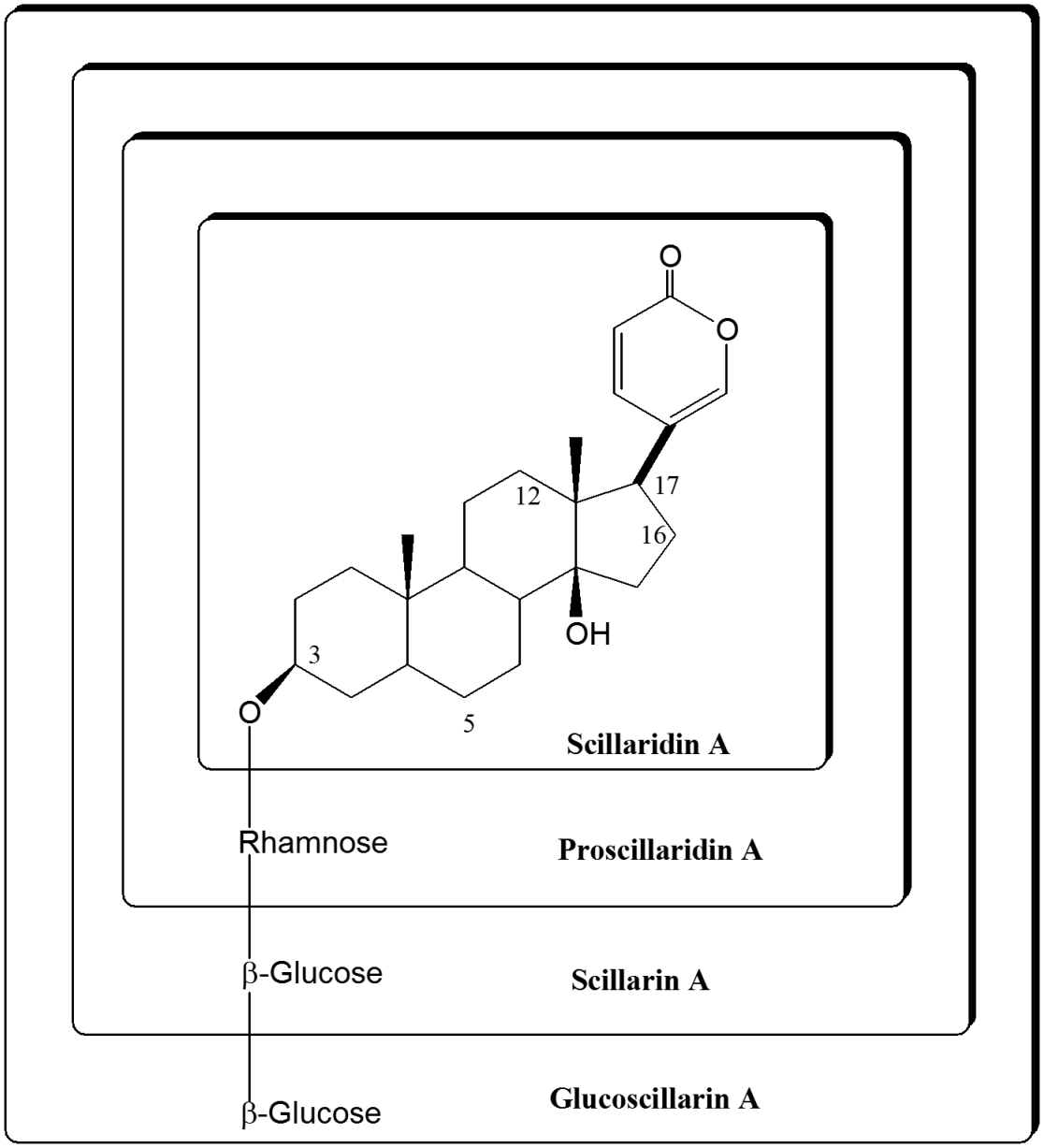
- glycosides are poorly absorbed in GIT → short action duration (not cumulative)
- In small doses: mild gastric irritation causing a reflex secretion of bronchioles
- Reflex expectorant action
- Larger doses: emetic
- Diuretic
- Cardiotonic

Cautions:

- Only under professional supervision.
- Toxic in excessive doses

Unofficial varieties

- Red Squill – contains squilloriside – very toxic
- Indian Squill – *Urginea indica*


O
O
12
17
16
OH
3
O
5
Scillaridin A
Rhamnose
Proscillaridin A
β-Glucose
Scillarin A
β-Glucose
Glucoscillarin A

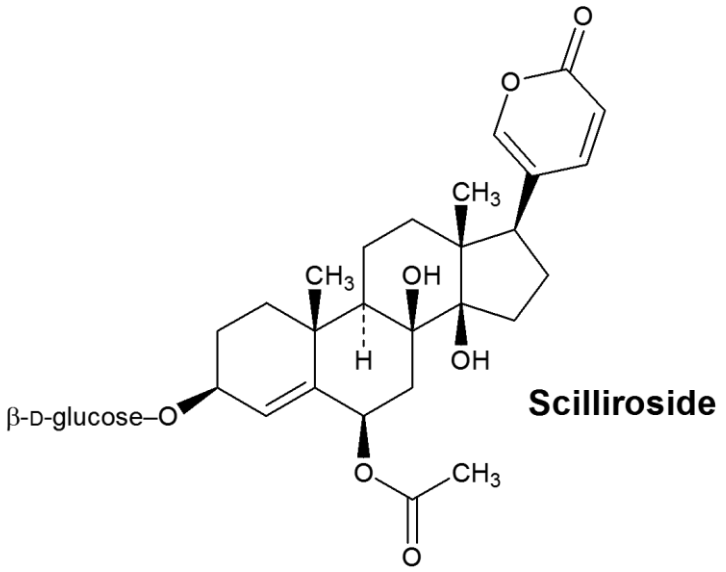
Pharmacological properties of squill

- **White squill:**

  it is an expectorant, but it also possesses emetic, cardiotonic (proscillaridin A), and diuretic properties.

- **Red squill:**

  it is used as a rat poison (scilliroside), because rodents lack the vomiting reflex, which makes red squill particularly lethal to these animals.



Scilliroside

(3β,6β)-6-(Acetyloxy)-3-(β-D-glucopyranosyloxy)-8,14-dihydroxybufa-4,20,22-trienolide

4. Simple Phenolic glycosides

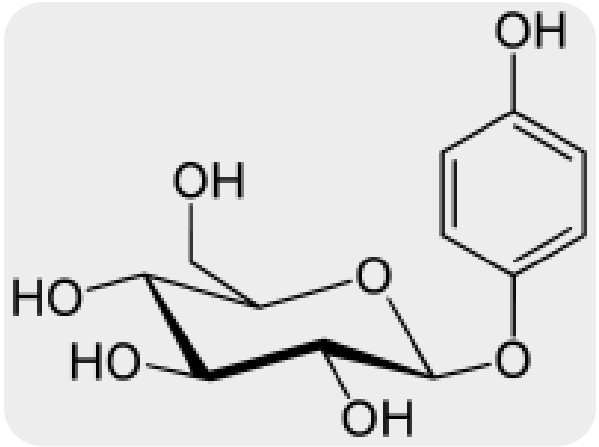
- These chemical structures are classified by the presence of a benzene ring combined with an alcohol.
- Most members of this class of compounds have antiseptic activity and are also anthelminitic and caustic.
- Other actions related to this class of compounds include fever reduction, general anti-inflammatory effect, increased volume and concentration of bile, and rubefacient actions when used externally.

- Anti-inflammatory effect is due to an anti-prostaglandin and bradykinin action, which would also explain the use of salicylates as anti-blood-clotting agents.
- In summary, these are generally useful compounds that are quite safe when used in their natural plant context.
- Two other familiar examples are
  - Eugenol from Cloves, which helps tooth aches, and
  - Thymol from Thyme and other aromatic remedies which is a useful anti-septic and anti-fungal oil.

Bearberry

- Synonyms: Busserole
- Biological source: It consists of dried leaves of *Arctostaphylous uva-ursi* Linn.
- Family: Ericaceae
- Geographical source: Europe, USA, canada & Scotland.
- Macroscopy:
- Colour - Olive green
- Odour - odourless
- Taste - Astringent and bitter

- Chemical constituents:
  - A phenolic glycoside arbutin
  - Also contains
    - methyl arbutin
    - Quercetin
    - Ursone
    - Quinones
    - tannins (6-10%) and
    - terpenoids



- Uses:
  - Diuretic
  - Astringent
  - As an infusion in urethritis and cystitis.

5. Cyanogenetic glycosides

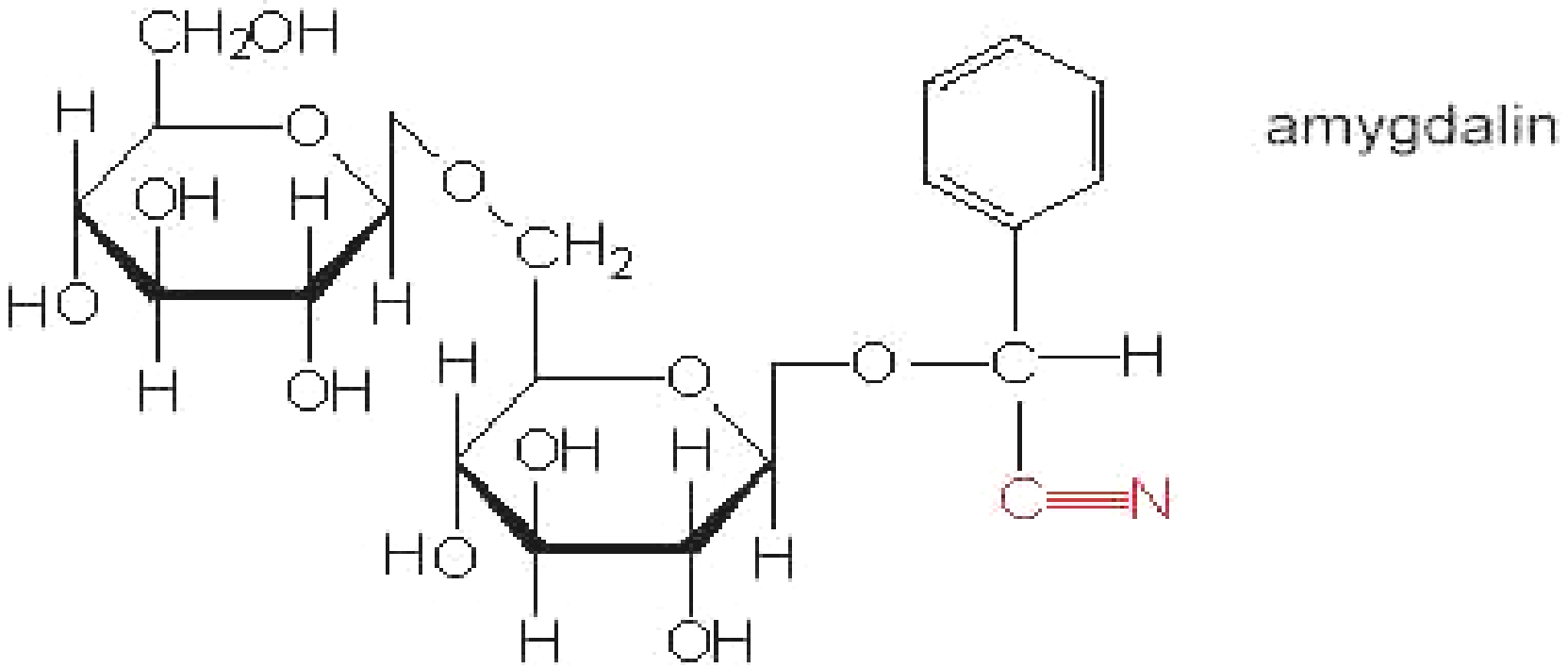
- Cyanogenesis is the ability of certain living organisms, plants in particular, to produce hydrocyanic acid (HCN, prussic acid) on hydrolysis.
- Cyanogenesis in plants is a chemical defense mechanism against organism damaging or feeding on plant tissues and lead to release of HCN gas, which is toxic.
- They are distributed in over 2000 plant species belonging to 110 families.
  - Examples such as Wild Cherry Bark, Almond

- These compounds, in presence of enzymes such as β-glucosidase, lose their sugar portion to form a cyanohydrin which, in the presence of water and hydroxynitrile lyase, can undergo hydrolysis to give benzaldehyde and the highly toxic hydrogen cyanide (HCN).
- The sugar portion of the molecule may be a monosaccharide or a disaccharide such as gentiobiose or vicianose. If a disaccharide, enzymes present in the plant may bring about hydrolysis in two stages, as in the case of amygdalin.
- Cyanogenic glycosides are not toxic by themselves

- Most cyanogenetic glycosides are biosynthetically derived from the amino acids
  - valine, leucine, isoleucine, tyrosine or phenylalanine or nicotinic acid.
- Cyanogenetic glycosides are easy to detect with a strip of filter paper impregnated with reagents able to give a colour reaction with the hydrocyanic acid released upon crushing the plant material
  - e.g., picric acid/sodium carbonate or benzidine/cupric acetate.

- Examples:

1. **<u>Amygdalin</u>** in bitter almonds (*Prunus amygdalus*). It is biosynthetically derived from phenylalanine.
2. **<u>Linamarin</u>** in linseed (*Linum usitatissimum*). It is biosynthetically derived from valine.



amygdalin

Bitter almond

- Synonyms: Amygdala amara
- Biological source:
  - Bitter almonds are dired ripe seeds of plants, *Prunus amygdalus*
- Family: Rosaceae
- Chemical constituents:
  - Glycoside known as amygdalin(1 to3%)
  - 40-50% fixed oil, 20% proteins, 0.5% Volatile oil
  - Enzyme emulsin
- Uses:
  - It is used as sedative due to hydrocyanic acid.
  - Demulcent
  - It is not used for flavouring foods.

Wild cherry bark

- Synonyms: Virginian prune bark
- Biological source:
  - It is the dried bark of *Prunus serotina*
- Family: Rosaceae
- Chemical constituents:
  - Glycoside called prunasin
  - Prunasin on hydrolysis gives benzaldehyde, glucose and hydrocyanic acid.
  - It also contains p-coumaric acid, benzoic acid, resin methyl aesculetin and trimethyl gallic acid.
- Uses:
  - It is used as expectorant and syrup is used as a flavouring agent.

6. Isothiocyanante glycosides

- These are glycosides based on the following general formula:
- The basic structure of glucosinolates comprises a glucose residue, a sulfate group found as potassium salt and a variable aglycone.
- They are common in the family Brassicaceae.
- Since glucosinolates are ionic compounds they can be separated on ion exchange resin.
- When the tissues of glucosinolates-containing plants are bruised, the glucosinolates are hydrolyzed by a thioglucosidase commonly known as myrosinase (myrosin) which is always found in this type of plants

6. Isothiocyanate glycosides

- Allyl isothiocyanate is strong-smelling volatile compound responsible for the astringent odour and flavour of mustard.
- Uses of allyl isothiocyanate
  - as condiment
- It is an irritant substance and therefore it is used as
  - Counter irritant (rubefacient)
    - Used in the form of plasters.
    - If the plasters are applied too long, they can produce skin lesions.

Black mustard

- Biological source:
  - These are the dried ripe seeds of *Brassica nigra* or *Brassica juncea*
- Family: Cruciferae
- Macroscopy:

Colour	-	Black or dark brown
Odour	-	None, crushed seeds have pungent odour
Taste	-	Bitter
Shape	-	spherical

- Chemical constituents:
  - Black mustard seeds contain about 4% of isothiocyanate glycoside called sinigrin.
  - Sinigrin on hydrolysis with water produce allyl isothipcyanate.
  - Fixed oil 30%
  - Proteins 20%
  - Volatile oil 0.7 to 1.3%
- Uses:
  - It is a condiment
  - It is emetic when used internally in higher doses
  - Externally it is used as counter irritant and rubefacient in the form of poultice or plasters.
  - All the activity of seed is due to sinigrin or consequent v.o